

Frontier AI for Actuaries.

AI helps actuaries perform complex, multi-step actuarial tasks with every step documented and reproducible.

THE WORKBENCH

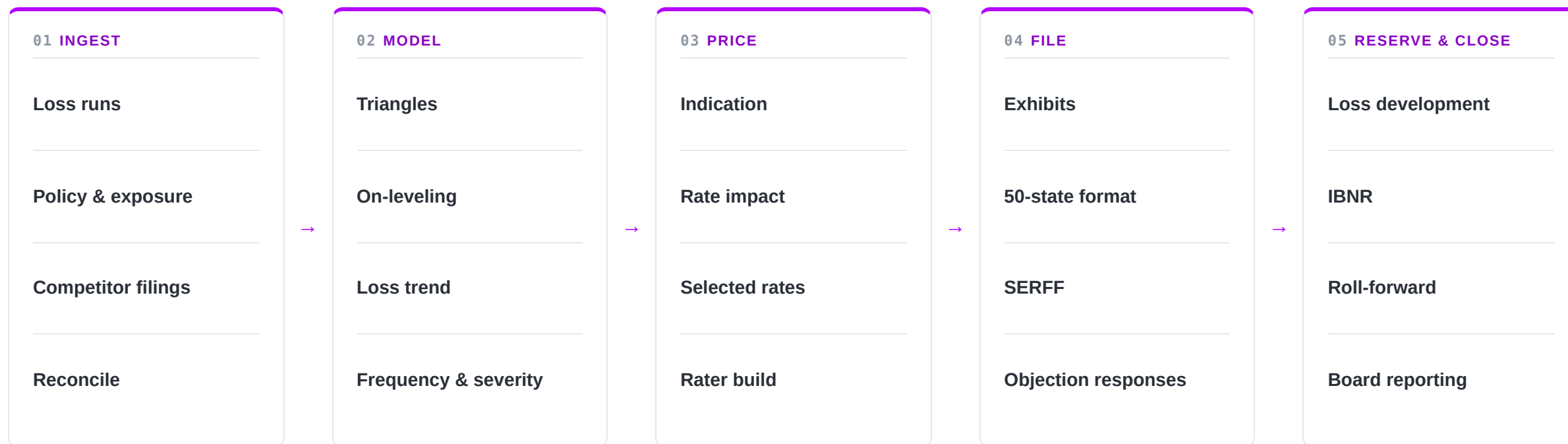
One collaborative workspace for pricing, reserving, and filing

For all actuaries at admitted and specialty carriers, MGAs,
and reinsurers.



Actuarial work is not one workflow. It is a **web of intricate, connected tasks**.

The more steps an actuary delegates to AI, the less certain they can be of the output. Tesora enables full tracing, documentation, and audit, so the actuary remains in control.



One agentic harness across all of it

A system of agents keeps every task linked and every value cited, from ingest to filing. That orchestration is the hard part to build, and the harder part to replicate.

AUDITED END TO END



A frontier model can already do the math...

...Tesora puts it in a format an actuary can **sign and file**.

A FRONTIER MODEL, ON ITS OWN **Confident, but guessing**

Claude

Here is my GA commercial auto paid-loss triangle, 2019 to 2025. Select development factors, fit a tail, and give me the indicated reserve and the on-leveled rate change.

C From a typical development pattern, age-to-age factors run around **1.55, 1.18, 1.08, 1.03**, with a tail near **1.02**, which puts ultimates a little above your latest diagonal. Assuming loss trend near 6% and standard on-leveling, a **mid-single-digit** rate change looks reasonable.

! WHY AN ACTUARY CANNOT SIGN THIS

- ! It never saw your triangle. A real book is millions of rows and will not fit in context.
- ! The factors and trend are generic, with no source to trace or reproduce.
- ! No on-leveling to your history, no exhibits, nothing built to ASOP.

THE SAME MODELS, INSIDE TESORA **Working your real filing**

Rate Rule Form

+ New

Retrieving trigger data for validation.
Show original

See previous prompt

> ⚡ Creating a filing record and verifying the status

THINKING
I'm uploading and registering all 16 files to the vault.

Uploading 16 files to Coalition (CIC) WA Rate Filing — rater review + competitor scan and creating the task tree.
Show original

> ⚡ Uploading documents and adding them to your vault

THINKING
I'm mapping the Coalition rating model and scanning the filing for mentions of competitors like Travelers and At-Bay. I'll need you to clarify the scope and provide data for any external market comparisons.

16 files received and processing. Starting the rating analysis.
Show original

> ⚡ Organizing tasks for analyzing insurance rates and competitors

Starting the rater task.
Show original

> ⚡ Starting a task and coordinating digital assistants



What's the best way to **implement AI**?

Your implementation of AI today will dictate its value, translatability, and reproducibility going forward.

	Actuarial consultants Milliman, WTW, Oliver Wyman	Your team with AI In-house, on ChatGPT or Claude	Actuarial software hyperexponential, Earnix, Akur8	T Tesora Frontier models, in your format
Turnaround	✗ Weeks to months, in their queue	≈ Quick drafts, still checked by hand	✓ Fast and repeatable once it is live	✓ Minutes to days, in your workflow
Setup & fit	≈ A fresh engagement scoped each time	✓ No one knows your book better	✗ Heavy implementation on their platform	✓ 60-day build in your VPC, on your own data
Your raters & filings	✗ Generic deliverables in their template	≈ Your format, rebuilt by hand each time	≈ Rebuilt inside their system, their way	✓ Your raters, versioned and ready to deploy
Source & audit	✓ Independent, credible workpapers	✗ No source to trace, and the answer drifts	≈ Logged inside the platform	✓ Cell-level citations, append-only log
Scope	≈ Whatever you scope and pay for	≈ One step at a time, gaps in between	≈ Pricing or reserving, rarely filing too	✓ Pricing, reserving, filing, and close in one place



Strength



Partial or conditional



Gap



Rate changes in **days**, not months.

Source a rater from a SERFF filing, an Excel file, or a script. Tesora rebuilds it, tests it, and deploys it whenever underwriting is ready.

BEFORE · BY HAND

4-6 months

From a requested change to a filed, live rater, and where the months actually go

- | | | |
|----|---|-------|
| 01 | Spec the change
Actuary writes up the revised rates and factors | DAYS |
| 02 | IT ticket, then the queue
The change waits for developer capacity | WEEKS |
| 03 | Re-code the rater
Engineering reimplements the logic by hand | WEEKS |
| 04 | Rebuild exhibits, 50 states
Each state reformatted by hand for SERFF | WEEKS |
| 05 | Manual QA
No test suite, so regressions are caught by eye | WEEKS |
| 06 | Compliance sign-off
Actuarial and legal review the filing | DAYS |
| 07 | File and answer objections
Back and forth with each department of insurance | WEEKS |
| 08 | Release window
Deploy only in the next scheduled window | WEEKS |

AFTER · ON TESORA

1-2 days

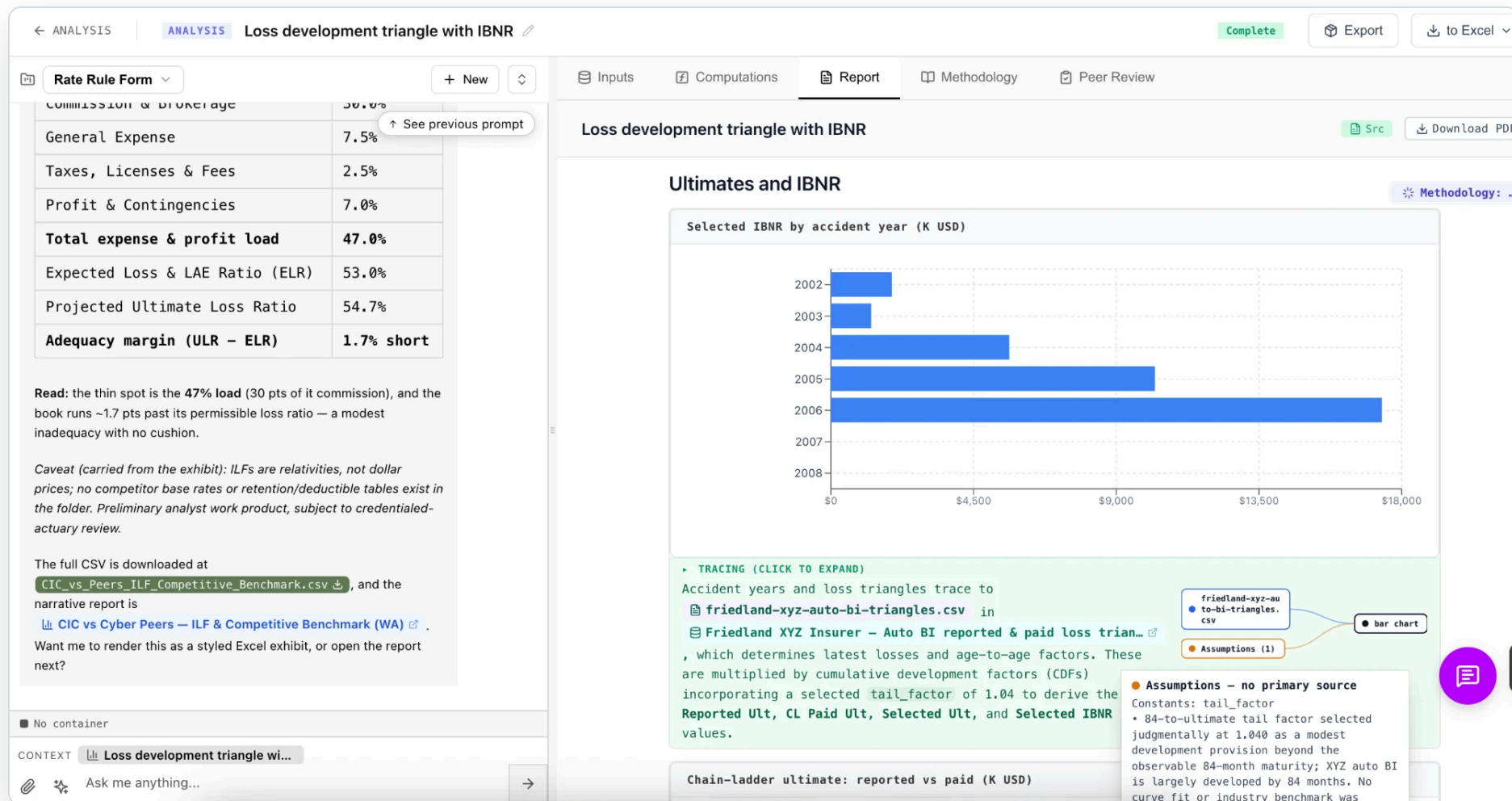
Deploy as a web interface or API.

- | | |
|----------|---|
| YOU GIVE | A plain-language change and the source rater, from SERFF, Excel, or a script. |
| YOU GET | A tested rater with a working GUI, sample ratings, and a versioned API. |



The triangle is squared before you sit down.

Reserving usually burns days pulling and reconciling triangles. Tesora does that on ingest, so your time goes to selecting factors.





Work you can reopen years later.

AI sessions save your work so you can revisit, reproduce, or audit the work many years later.

Tesora

Home

Work Tracking

ARTIFACTS

Folders

Reports

Raters

Routines

APPS

Compute St...

Underwriting

Rater Recon...

Competitive ...

GLM Studio

Triangle Buil...

All Apps

Support

Settings

vivek

Back

EXAMPLE

Bernegger 2024 — Integrated experience/exposure rating (ASTIN)

EX-VLT-bernegger-2024-integrated-rating

Analyze Data

Build

Rate Rule Form

+ New

See previous prompt

Profit & Contingencies	7.0%
Total expense & profit load	47.0%
Expected Loss & LAE Ratio (ELR)	53.0%
Projected Ultimate Loss Ratio	54.7%
Adequacy margin (ULR - ELR)	1.7% short

Read: the thin spot is the 47% load (30 pts of it commission), and the book runs ~1.7 pts past its permissible loss ratio — a modest inadequacy with no cushion.

Caveat (carried from the exhibit): ILFs are relativities, not dollar prices; no competitor base rates or retention/deductible tables exist in the folder. Preliminary analyst work product, subject to credentialed-actuary review.

The full CSV is downloaded at `CIC_vs_Peers_ILF_Competitive_Benchmark.csv`, and the narrative report is [CIC vs Cyber Peers — ILF & Competitive Benchmark ...](#). Want me to render this as a styled Excel exhibit, or open the report next?

No container

CONTEXT Bernegger 2024 — Integrated...

Ask me anything...

Back to documents

bernegger-2024-integrated-experience-exposure-rating.pdf

Download

Origin

Type

Size

Status

Uploaded

Document ID

File ID

User upload

application/pdf

1.5 MB

pending

Aug 7, 2026, 06:57 PM

D0C-5c0c8dfe-1a3a-4eeb-8c76-5ab1947ac4dd

54238c79-b072-45c7-92cb-096cfb81f242

SUMMARY

No summary available. The ingestion workflow either hasn't finished or didn't produce one for this document.

1 / 28

Generic framework for a coherent integration of experience and exposure rating in reinsurance

Stefan Bernegger

Independent Scholar, Goldenerstrasse 7B, CH-8127, Fench, Switzerland

Email: st.bernegger@gmail.com

Received: 3 October 2023; Revised: 11 April 2024; Accepted: 12 April 2024; First published online: 27 August 2024

Keywords: Experience rating; exposure rating; short-tail; long-tail; generative model; Bayesian inference; reduced statistics; model calibration; Monte Carlo simulation

Abstract

This article introduces a comprehensive framework that effectively combines *experience rating* and *exposure rating* approaches in reinsurance for both *short-tail* and *long-tail* businesses. The generic framework applies to all non-life lines of business and products emphasizing nonproportional treaty business. The approach is based on three pillars that enable a coherent usage of all available information. The first pillar comprises an exposure-based *generative model* that emulates the *generative process* leading to the observed claims experience. The second pillar encompasses a standardized *reduction procedure* that maps each high-dimensional claim object to a few weakly coupled *reduced random variables*. The third pillar comprises calibrating the generative model with retrospective *Bayesian inference*. The derived calibration parameters are fed back into the generative model, and the reinsurance contracts covering future cover periods are rated by projecting the calibrated generative model to the cover period and applying the future contract terms.

1. Introduction

There are two approaches to probability theory: the first is the *frequentist* interpretation, also known as *standard*, *orthodox*, or *classical* statistics, and the second is the *Bayesian* interpretation (Daykin *et al.*, 1994; Lee, 2012; Gelman, 2013; Klugman *et al.*, 2019; Murphy, 2022, 2023). This distinction also applies to the various statistical and modeling approaches.

The Bayesian interpretation has multiple advantages, but frequentist approaches are still popular and widely used for two main reasons. First, they are computationally ‘cheap’ compared to Bayesian methods, and second, they do not presuppose knowledge regarding prior. Bayesian inference is, nevertheless, widely applied in various domains of science (Parr *et al.*, 2022; Palmer, 2022; Harrison *et al.*, 2023), and it has become the foundation of machine learning (Murphy, 2022, 2023). Bayesian approaches are also gaining popularity in disciplines like insurance and reinsurance, which have traditionally been dominated by frequentist approaches (Bühlmann and Straub, 1970; Bühlmann and Gisler, 2005; Zhang *et al.*, 2012; Klugman *et al.*, 2019; Goffard *et al.*, 2023).

Reinsurance companies face multiple challenges when using models for assessing the assumed risks and for taking business decisions like risk selection, transactional pricing, reserving, investing, and corporate risk management, and they rely on a variety of methods and tools for managing these risks (Antal, 2009; Albrecher, 2017; Mildenhall, 2022). The actuarial models implemented by a reinsurer are based on simplified assumptions, and reinsurers know that the underlying assumptions are rarely fulfilled in the real world. One way to deal with the issue is to complement experience-based rating models with exposure-based models. The latter can incorporate all relevant features of the generative process that affect the rating. All such features must be calibrated, which is accomplished with market data. However, cedent-specific features (hidden variables) must also be reflected in the rating of reinsurance contracts.

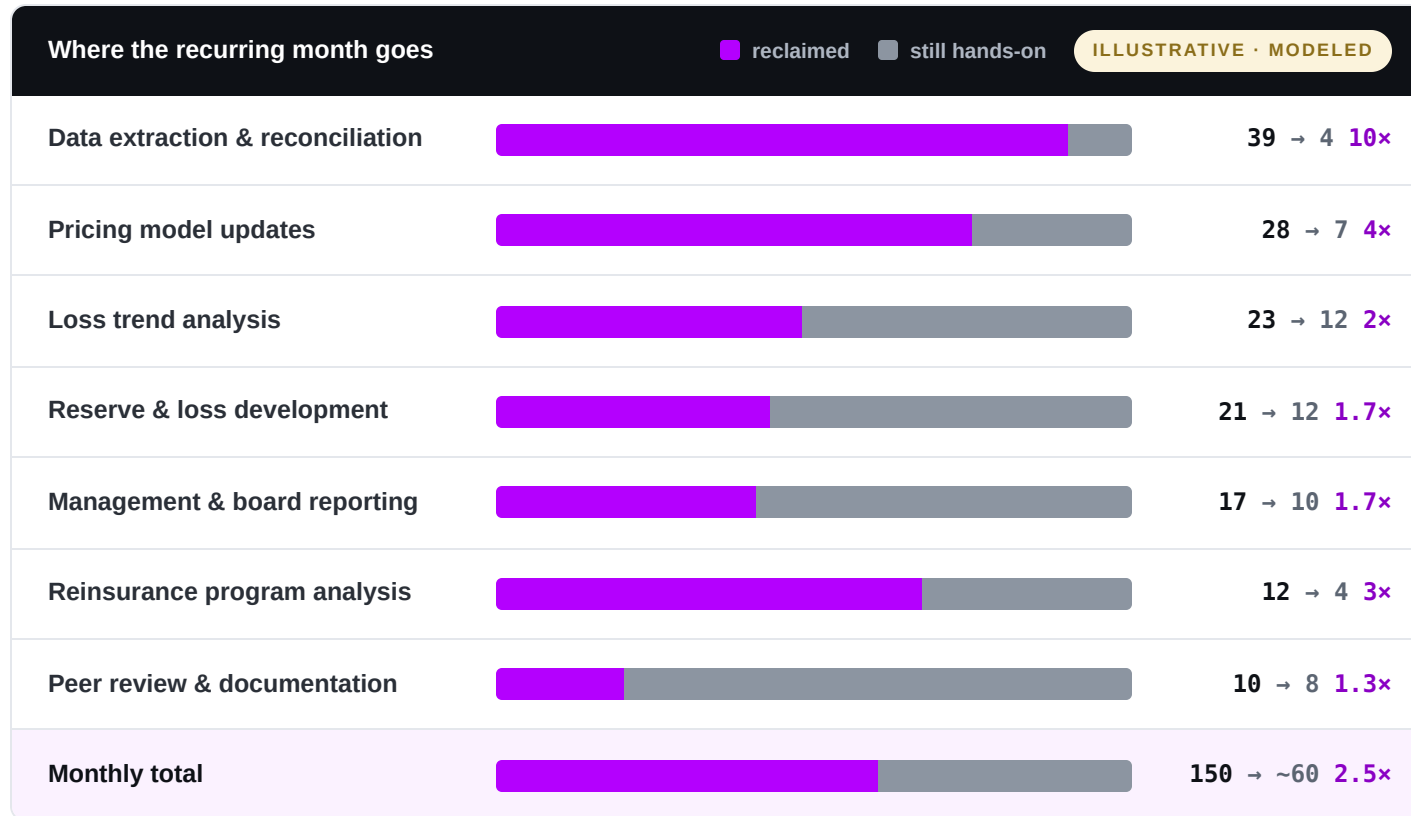
© The Author(s), 2024. Published by Cambridge University Press on behalf of The International Actuarial Association. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted re-use, distribution and reproduction, provided the original article is properly cited.

<https://doi.org/10.1017/ins.2024.17> Published online by Cambridge University Press



The recurring month goes from 150 hours to about 60.

Tesora compresses the mechanical grind, the data pulls, the reconciliations, the model updates, so the same team spends its hours on judgment instead of assembly.



2.5×

the recurring workload, same team and sign-off

The blended throughput gain across the whole recurring month.

Smarter not harder

Time freed for strategy and selections, not assembly

Unlock your actuaries

More products, territories, and actuarial capability

Hours reflect a typical ~150 hour recurring actuarial month, with per-workflow gains from Tesora's engagement framework. Figures are modeled and confirmed against measured results during a build.



It makes us **even better.**

Better models accelerate the product underneath you, compounding your advantage with every release.



Every release lifts you

The workbench is model-agnostic and runs on the strongest model available. Each new generation handles your schemas and messy edge cases more easily, so the work gets sharper **at no cost to you.**



We keep learning your systems

The longer we work with your team, the more of your data, formats, and quirks the workbench encodes, and that knowledge carries across every model upgrade.



The cheapest correct path

Every org is running a dozen AI pilots. We stay anchored on one thing: the fastest, most correct, and lowest-token way to run a complex workflow within your audit standards.



Audits matter more, not less

As the model does more of the work, proving it matters more, so no step gets skipped and nothing is taken for granted. Every value stays cited, reproducible, and signed off.



The gap widens

Better models unlock even more complex work, so the advantage compounds. A rate change that took months now takes days, so you can reprice against the market while slower competitors wait on a filing cycle. The firms that move first pull ahead, and **standing still gets more expensive.**



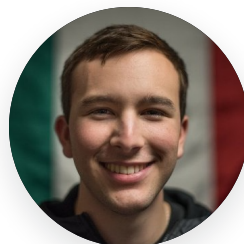
A team built at the intersection of frontier AI research and decades of actuarial experience, **the two disciplines this problem demands.**



Vivek Rao

CEO

Former private-equity investor and McKinsey consultant focused on P&C insurance.



Federico Reyes

CTO

Stanford AI researcher. Graph neural nets and relational data at Kumo AI (acquired by Nvidia), and on Google's Document OCR team.



Philo Bishay

CPO & CHIEF ACTUARY

Chief actuary at W. R. Berkley E&S, owning pricing and reserving. Capital modeler at **Gen Re**; built and sold a brokerage.

BACKED BY



Foundation Capital

Leading AI and enterprise investors since 1995



Y Combinator

S25

The accelerator behind Stripe and Airbnb

One team, focused on solving this for every actuary

Above all, we want to see you win.

We are a thought partner first. This technology is powerful, and it only helps once it is opinionated: built with a real point of view on how the work gets done and defended.

We are here to push the frontier forward with the right partners, not to sell you a license. Doing right by you comes first, so every deployment starts with a proof of concept anchored on your organization's own success metrics. Some teams are stronger at reserving, others at pricing, and we work with yours to find where the workbench sets the best practices and shared learnings first.

[Start a POC discussion](#)

sales@tesora.ai

DE-RISKED BY DESIGN

- ✓ It runs in your own environment. We never hold your data.
- ✓ SOC 2 Type II, with the highest data-security controls.
- ✓ We clear your security review before any work touches a filing.